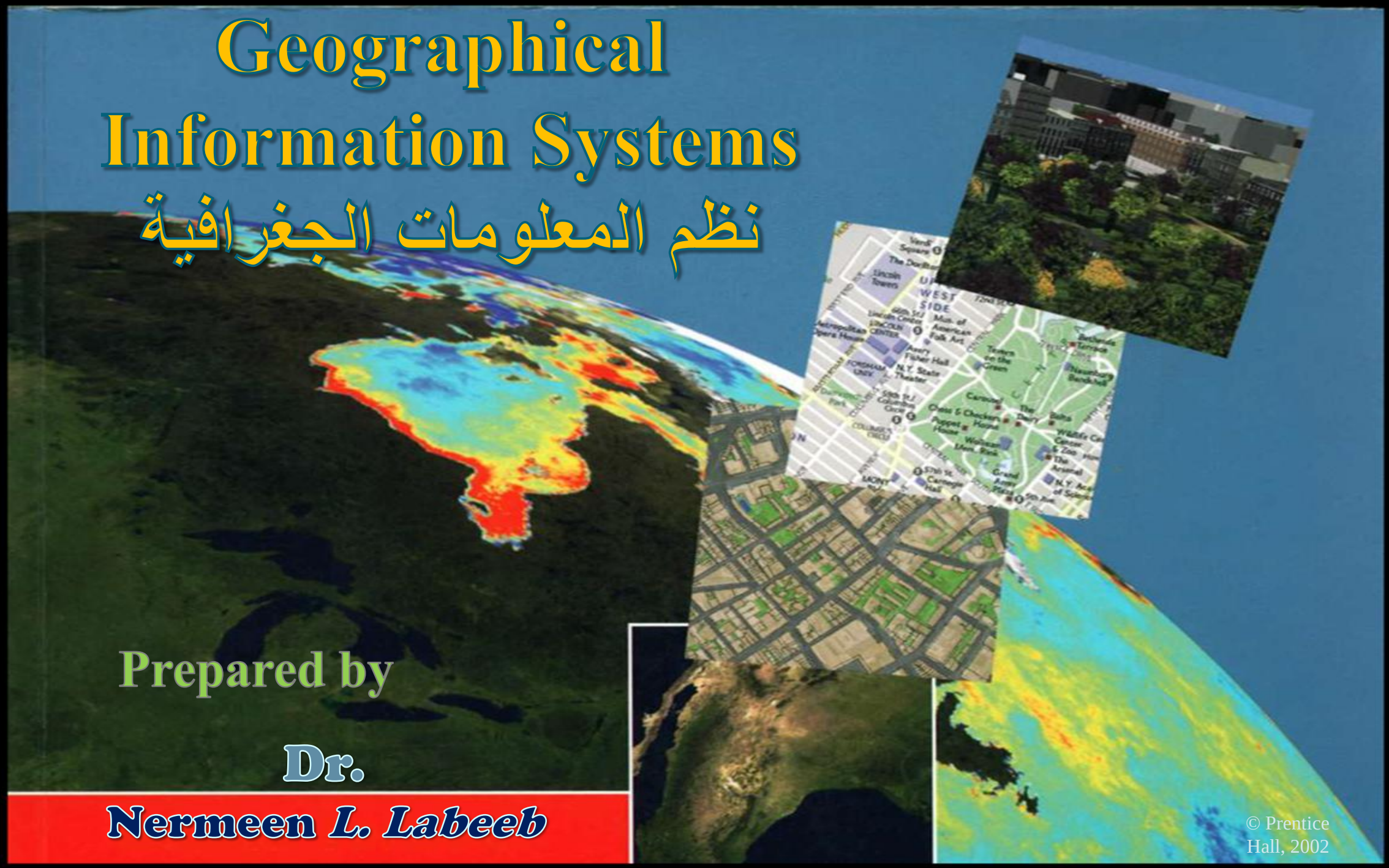
Geographical Information Systems

نظم المعلومات الجغرافية



Prepared by

Dr.

Nermeen L. Labeeb

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PART 1 Fundamentals of GIS

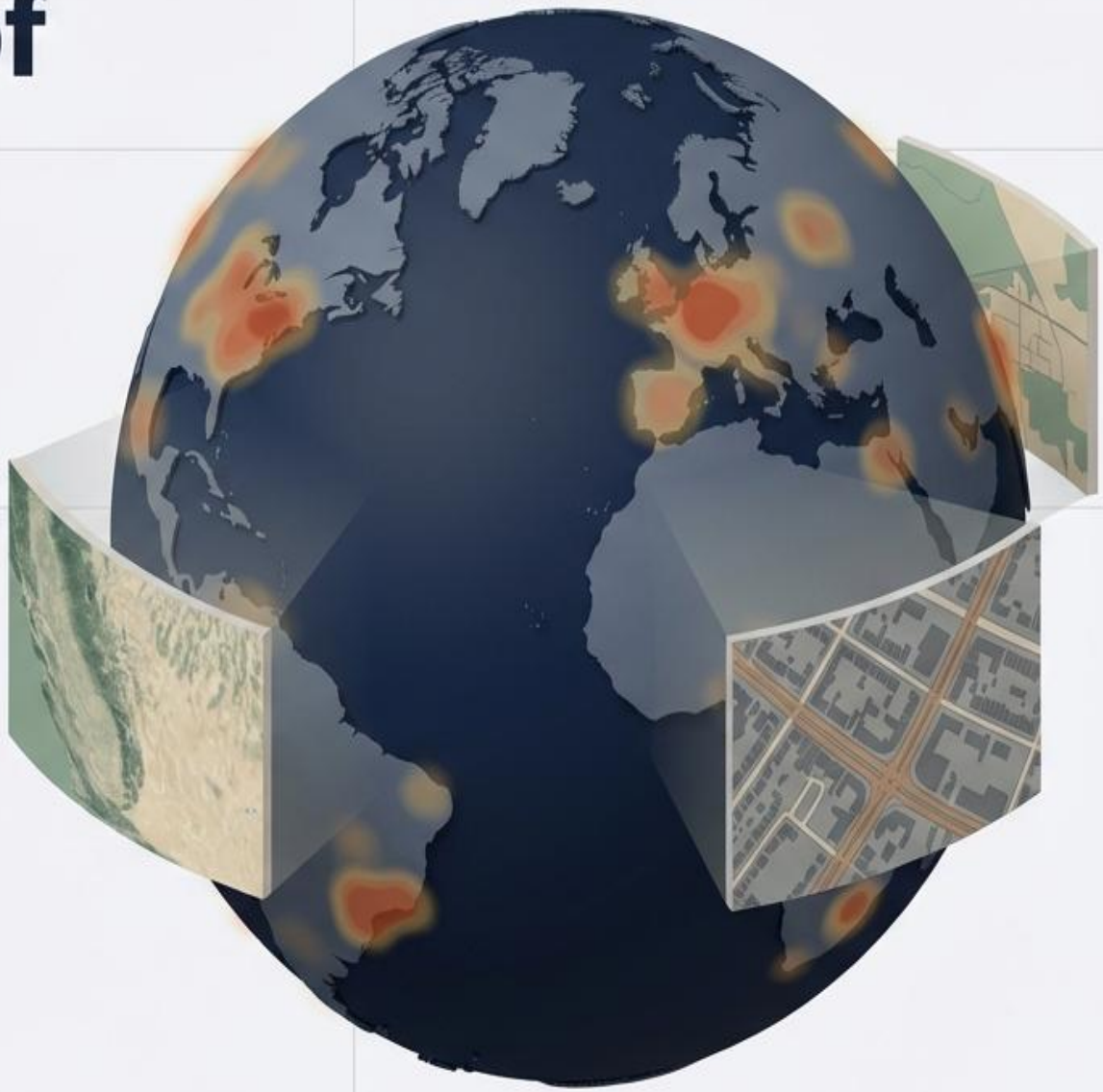
CHAPTER 1

What is GIS?

By the end of this chapter you should be able to:

- Explain what **GIS** is.
- Outline the **components of GIS**.
- Give examples of the **applications** of **GIS**.
- Describe how the **real world** is represented in **GIS**.
- Provide examples of the **analysis GIS** can perform.

Fundamentals of Geographic Information Systems



Managing and Understanding the World Through Spatial Data

1.1 INTRODUCTION

- Every day you ask questions with a ***spatial*** component.
- ***GIS*** has particular value when you need to answer questions about:

1. **Location**: Where is the **nearest** bookshop? Where are areas of forestry **in which state** can be found? Where is a certain restaurant can be found **in my area**?

2. **Patterns**: Where do high **concentrations** of students live **in this city?** What is the flow of traffic **along this motorway**? What is the **distribution of crime incidents** in London?

3. **Trends**. How are patterns of retailing changing in response to the development of **out-of-town superstores**?

4. **Conditions**. Where can I find holiday accommodation that is **within 1 km of** a *wind surfing* beach and *accessible by public transport*? Where is there flat land **within 500 m of** a *major highway*? Where are there over 100,000 *potential customers* **within a 5-mile radius** of a *railway station*?

5. **<u>Implications</u>**. If I move to a new home in this location, **how far** will I be *from the office, gym or coffee shop*? If we build a new park here, **what will be the effect on traffic flows**? What would be **the time saving** if we delivered our parcels **using this route, rather than an alternative**?

- The **generic questions** that a GIS can help to answer can be summarized as:
  - Where are ***particular features*** found?
  - What ***geographical patterns*** exist?
  - Where have ***changes occurred*** over a ***given period***?
  - Where do ***certain conditions apply***?
  - What will the ***spatial implications*** be if an organization takes certain action?

- **Case study(1)** provides examples of problems that can be addressed by asking these types of questions in an imaginary **ski resort** called **Happy Valley**.

questions with a spatial component

GIS provides the framework to answer five specific categories of inquiry about our environment:

1. Location

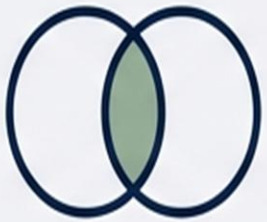
Where is it?
(e.g., Where is the nearest bookshop? Where can a certain restaurant be found?)

2. Patterns

Where are the concentrations?
(e.g., Where do students live? What is the distribution of crime incidents?)

3. Trends

How has it changed?
(e.g., How are retailing patterns changing in response to out-of-town superstores?)



4. Conditions

Where do factors overlap?
(e.g., Finding accommodation within 1km of a beach and accessible by transit.)

5. Implications

What happens if...?
(e.g., If we build a new park here, what will be the effect on traffic flows?)

1.2 DEFINING GIS

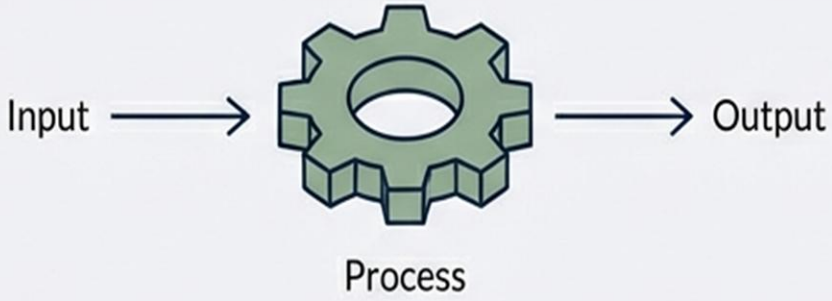
- A ***System*** : a set of ***interrelated components*** working together to achieve a specific ***goal*** by accepting ***inputs*** and ***processing*** them to produce ***outputs***.

- An ***Information System***: a set of ***interrelated components*** (**hardware**, **software**, **people**, **communications networks**, and **data resources**) that
  - *collect*, *process*, *store*, and *distribute* ***information*** to support
    - *decision making*, *coordination*, *control*, *analysis*, and *visualization* in an organization.

So,
What about GIS?

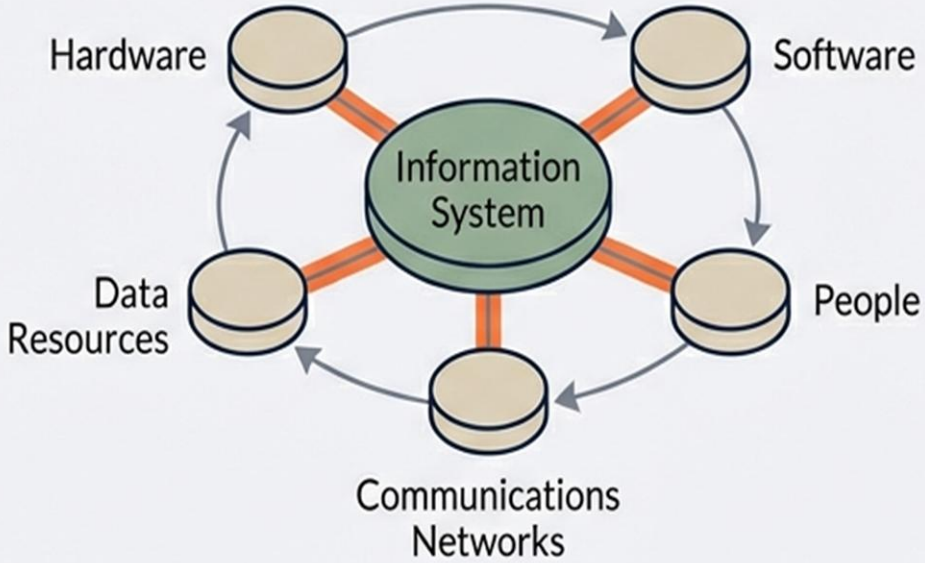
Constructing the Definition: From System to Information.

A System



A set of interrelated components working together to achieve a specific goal.

An Information System



Purpose: To collect, process, store, and distribute information to support decision making, coordination, control, analysis, and visualization.

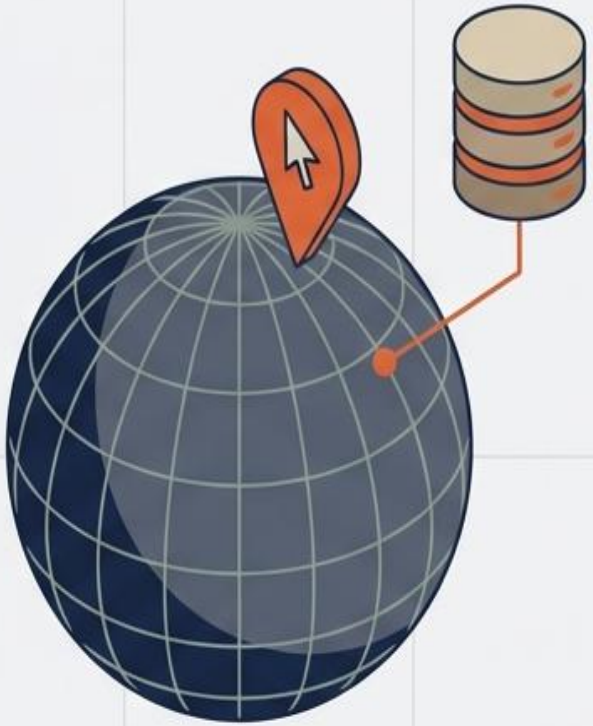
What is a GIS?

- A GIS is a tool for making and using ***spatial information***.
- It is a ***computer-based system*** to aid in the *collection*, *maintenance*, *storage*, *analysis*, *output*, and *distribution* of ***spatial data and information*** which are ***spatially referenced*** to the Earth.
- This is normally involves :
  1. ***Spatially referenced*** computer ***database***.
  2. Appropriate ***application software***.

- In short, GIS can be used to ***add value to spatial data***.
- GIS creates useful information to help decision making by:
  - Allowing ***data*** to be **organized** and **viewed** efficiently,
  - Integrating them with other data,
  - Analysis, and
  - Creation of new data that can be operated on in turn.
- In this way, a ***GIS*** can be described as a form of ***spatial decision support system.***

What is a GIS?

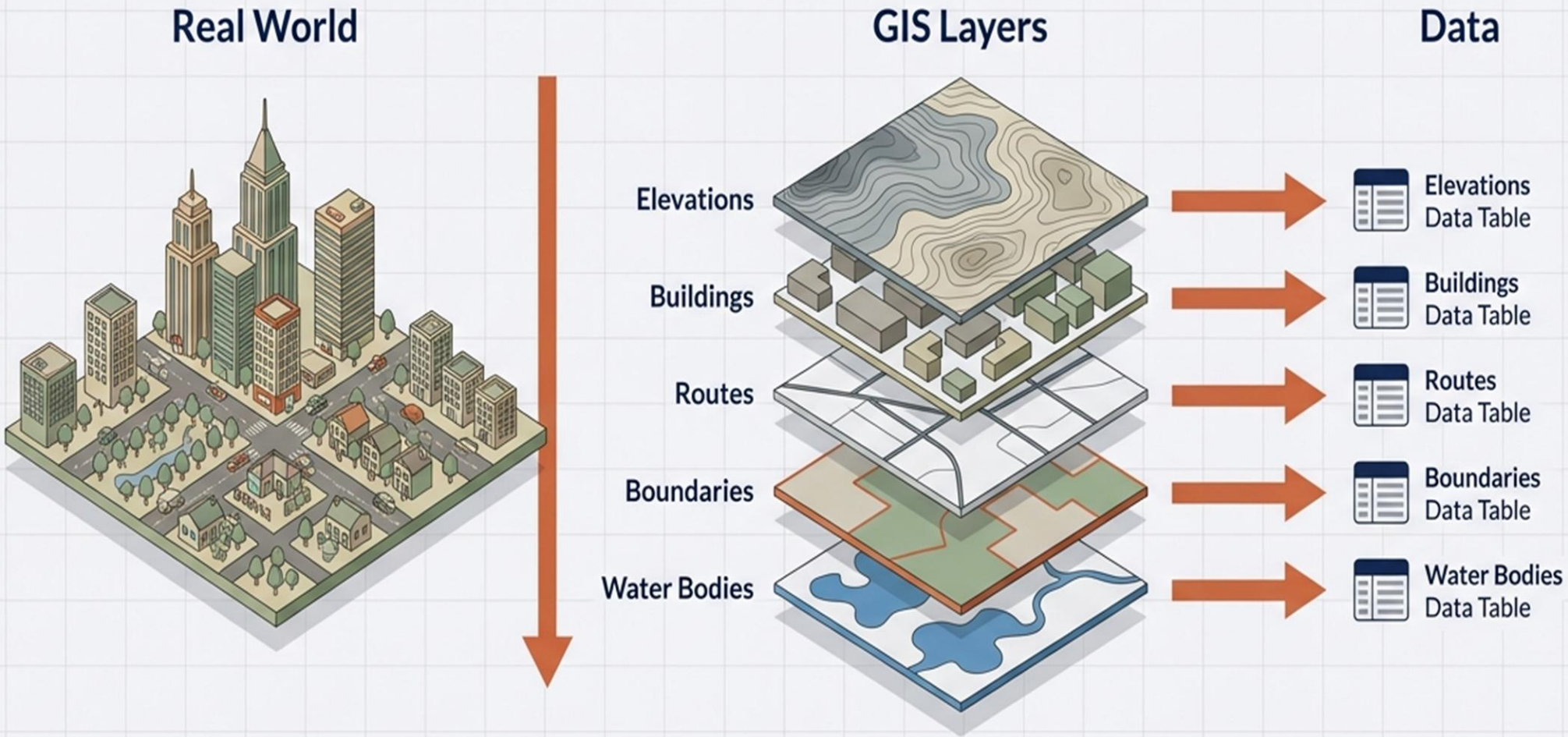
A computer-based system designed to aid in the collection, maintenance, storage, analysis, output, and distribution of spatial data and information.



Key Differentiator: All data is spatially referenced to the Earth.

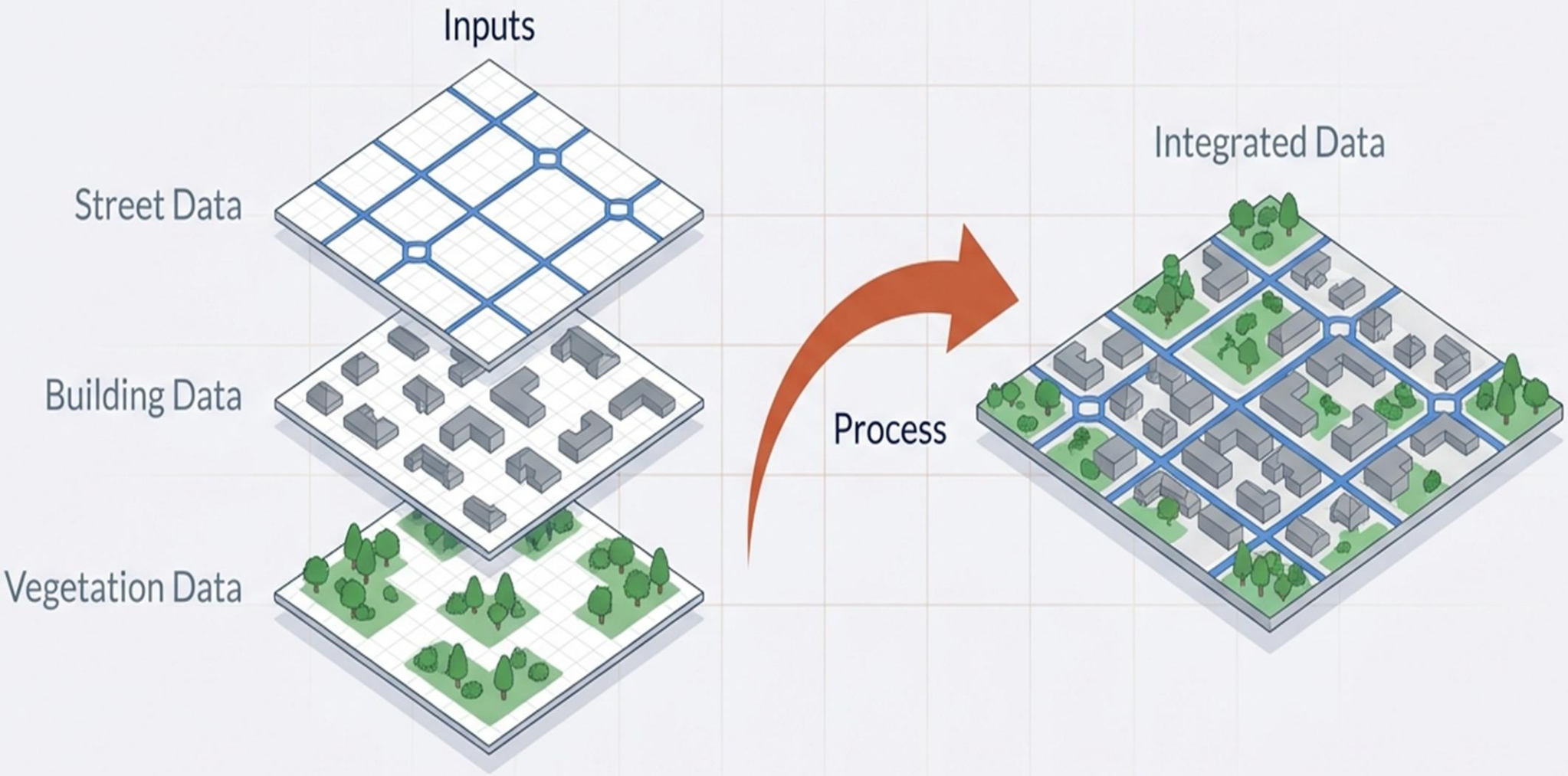
Requirements:
1. Spatially referenced computer database. 2. Appropriate application software.

Representing Reality Through Layers



GIS breaks down the real world into thematic layers that can be analyzed independently or stacked to form a complete picture.

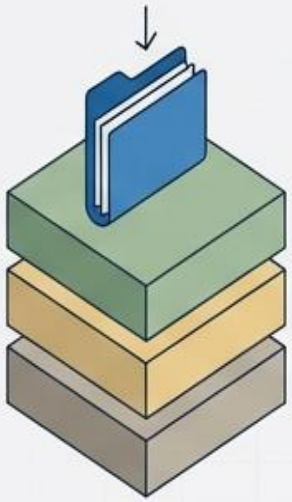
Integrated Data: The Power of Combination



By stacking distinct datasets, we can understand relationships that aren't visible in isolation.

Adding Value to Spatial Data

GIS creates useful information to support decision-making through four key functions:

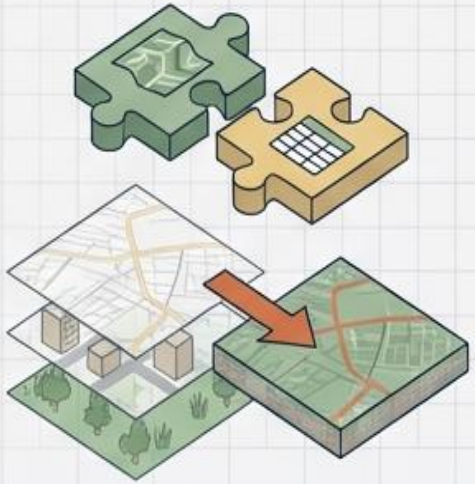


Organization

Allowing data to be organized and viewed efficiently.

Integration

Combining spatial data with other data types.

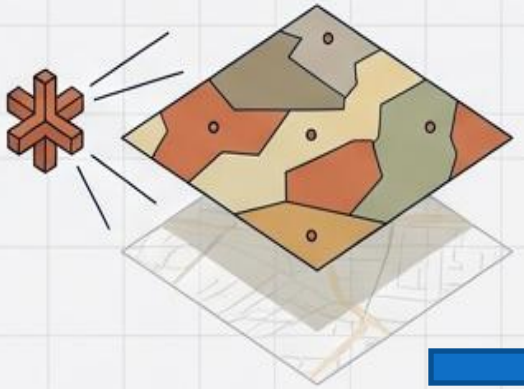


Analysis

Examining spatial relationships and patterns.

Creation

Generating new data from existing sources.



GIS is a Spatial Decision Support System

The ultimate goal of collecting, processing, and storing spatial data is to support organizational action.

- Decision Making
- Coordination
- Control
- Analysis
- Visualization

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